

Update on the management of postoperative nausea and vomiting.



New antiemetic drug developments, formulations, guidelines, risk evaluation, and controversies have occurred in the area of postoperative nausea and vomiting (PONV). These developments have helped improve our understanding of the prevention and treatment of PONV in the postanesthesia care unit and after discharge home or to the hospital ward. Antiemetic drug research has resulted in the introduction of the second-generation 5-hydroxytryptamine-3 (5-HT₃) receptor antagonist palonosetron and the neurokinin-1 (NK-1) receptor antagonist aprepitant, as well as new data on existing antiemetics. The next frontier and need for further nausea and vomiting research and therapy is the area of postdischarge nausea and vomiting after the patient is discharged home from phase II of the ambulatory stepdown unit or to the hospital ward. Antiemetic drug selection depends on efficacy, cost, safety, and ease of dosing. Safety concerns have arisen regarding the side effects of antiemetics, specifically their effect on the ECG with prolongation of the QTc interval by the butyrophenones and the first-generation 5-HT₃ receptor antagonist class of antiemetics. The impact of pharmacogenetics on antiemetic drug metabolism and their resulting efficacy has been correlated with genetic makeup affecting drug response. A discussion of ethics in PONV research has been initiated by the meta-analysis of PONV studies. To help guide antiemetic selection and PONV therapy for clinical practitioners, the Society of Ambulatory Anesthesia (SAMBA) PONV consensus guidelines have been introduced and updated.